



Registration of 3D CT Data

Alignment of Pre- and Post-Deformation Sandstone CT Volumes

Dataset: VEC4-bin2 | $380 \times 275 \times 275$ voxels | 16-bit CT

The Problem

Two CT Scans

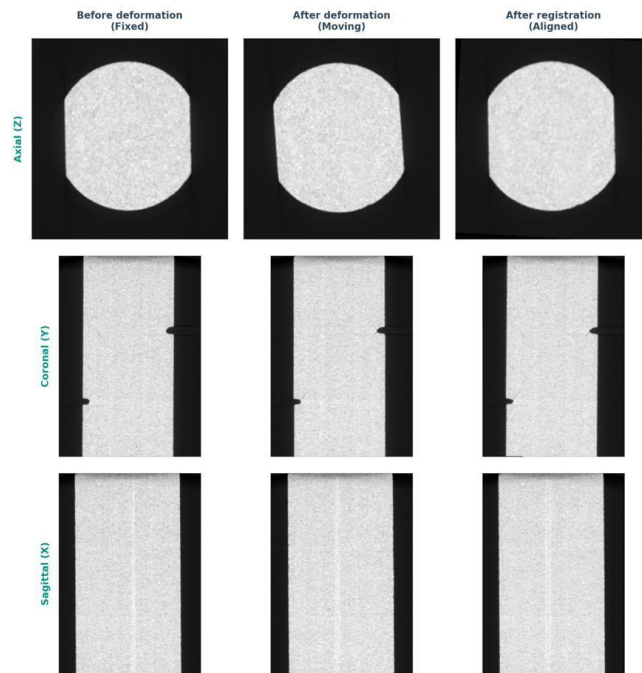
The same sandstone block scanned before and after mechanical deformation.

Repositioning Error

Between scans, the block was removed and placed back introducing an unknown rotation and shift.

Goal

Correct the repositioning error (rigid registration), then compare the volumes to detect real deformation.



Registration Approach



Rigid Transform

Euler3D - 6 parameters
(3 rotations + 3 translations)
Correct model for a solid object



Mutual Information

Mattes MI (50 bins)
Robust to intensity differences
between scan sessions



Multi-Resolution

3-level pyramid ($4\times \rightarrow 2\times \rightarrow 1\times$)
Coarse-to-fine prevents
local minima



Gradient Descent

Regular Step optimiser
Relaxation factor 0.5
500 iterations per level

Full Algorithm & Technical Details

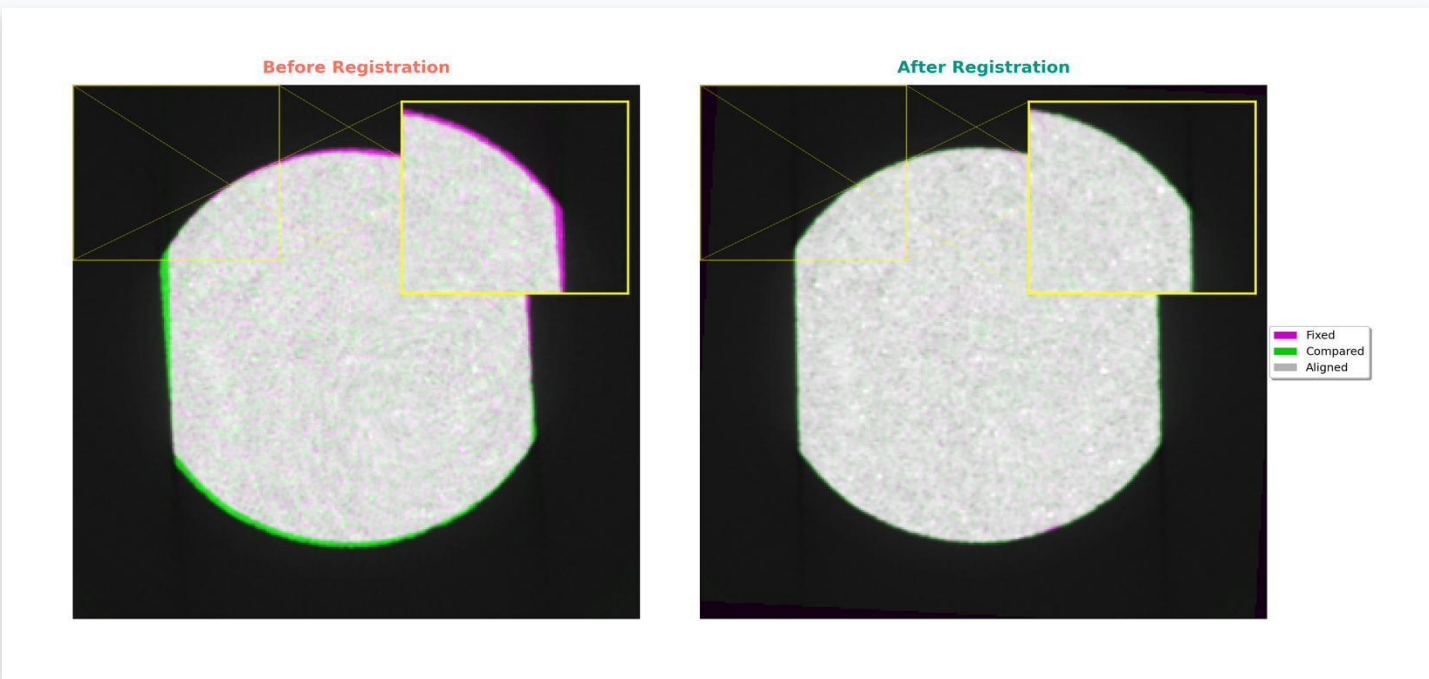


Parameter	Value
Transform model	Euler3DTransform (rigid: 3 rotations + 3 translations)
Initialisation	Centre-of-mass (MOMENTS)
Similarity metric	Mattes Mutual Information (50 histogram bins)
Metric sampling	RANDOM, 15 % of voxels per evaluation
Optimiser	Regular Step Gradient Descent
Learning rate	1.0 (initial)
Min step size	1×10^{-4}
Relaxation factor	0.5
Max iterations/level	500
Gradient tolerance	1×10^{-6}
Scale estimation	Physical shift (automatic)
Multi-resolution	3 levels
Shrink factors	4 → 2 → 1
Smoothing sigmas	2.0 → 1.0 → 0.0 (physical units)
Interpolator	Linear (registration) / Linear (resampling)
Preprocessing	Cast to float32, isotropic 1×1×1 spacing
Mask (metrics)	Otsu threshold + morphological cleanup; eroded 5 px

Tools: Python 3 | SimpleITK (registration + resampling) | NumPy + SciPy (masking, metrics) | scikit-image (SSIM) | Matplotlib (visualisation) | tiff file (I/O)

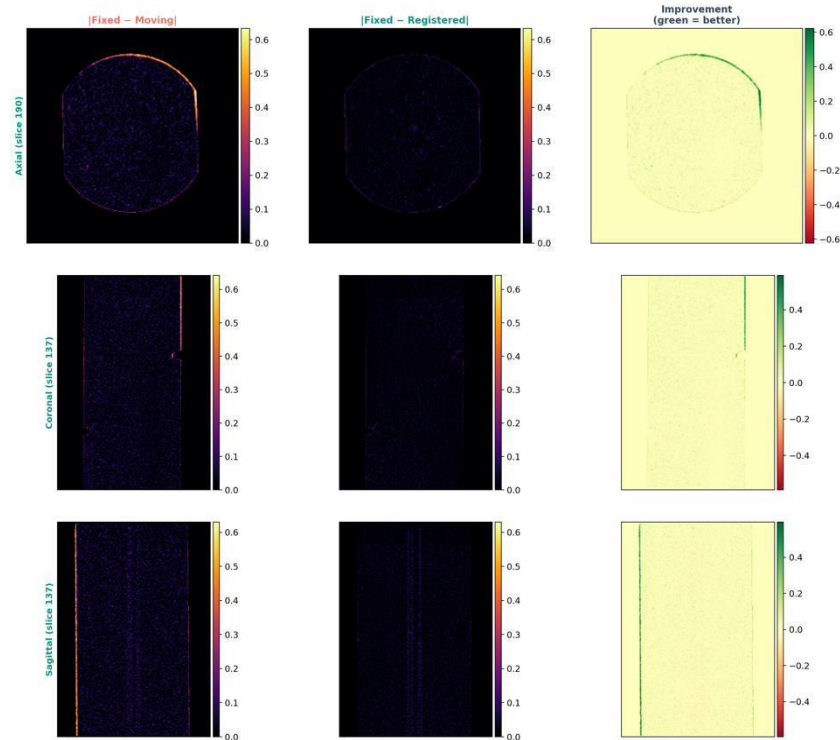
Visual Proof: Misalignment Comparison

- Axial slice



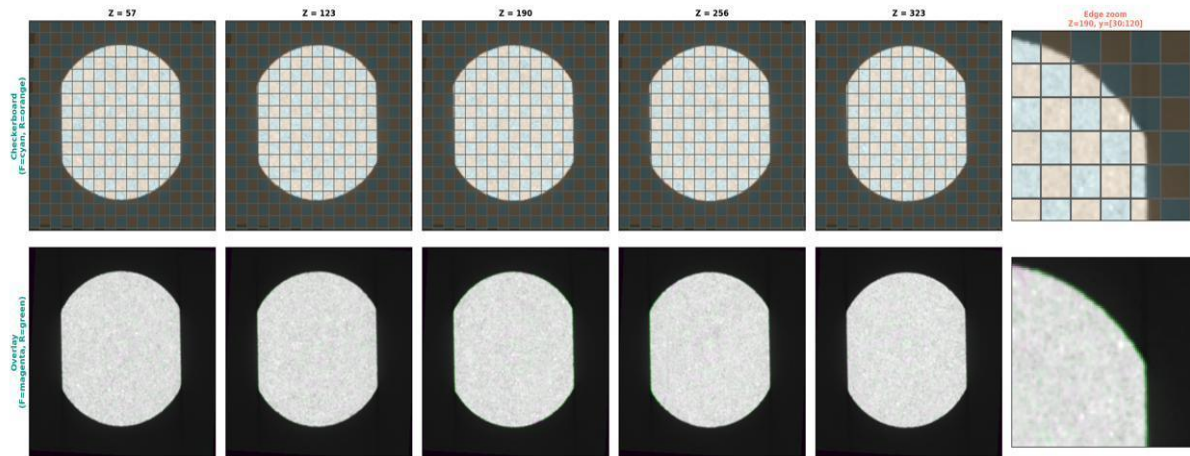
Difference Maps: Before vs After Registration

- Masked to specimen only - bright pixels indicate large differences



Checkerboard Verification

- Alternating 16px tiles from Fixed (cyan) and Registered (orange) - smooth transitions confirm good alignment



Quantitative Registration: Quality and Results

87%

MSE Reduction

0.00448 → 0.00058

0.997

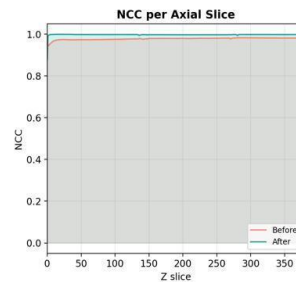
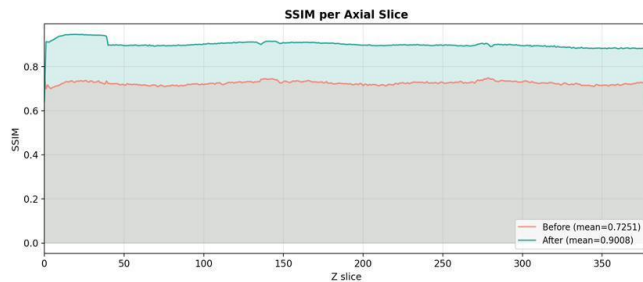
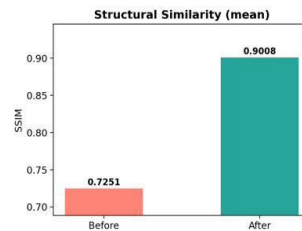
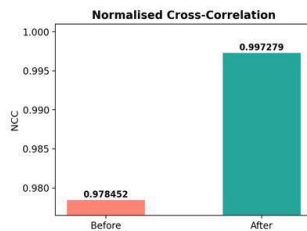
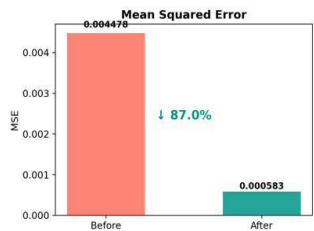
NCC After

up from 0.978

0.901

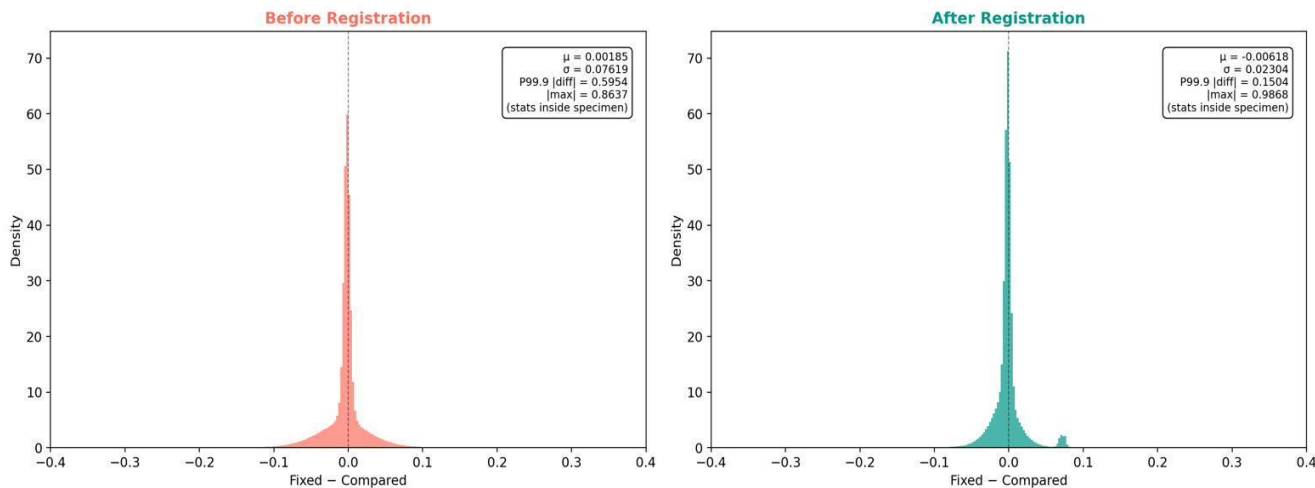
SSIM After

up from 0.725

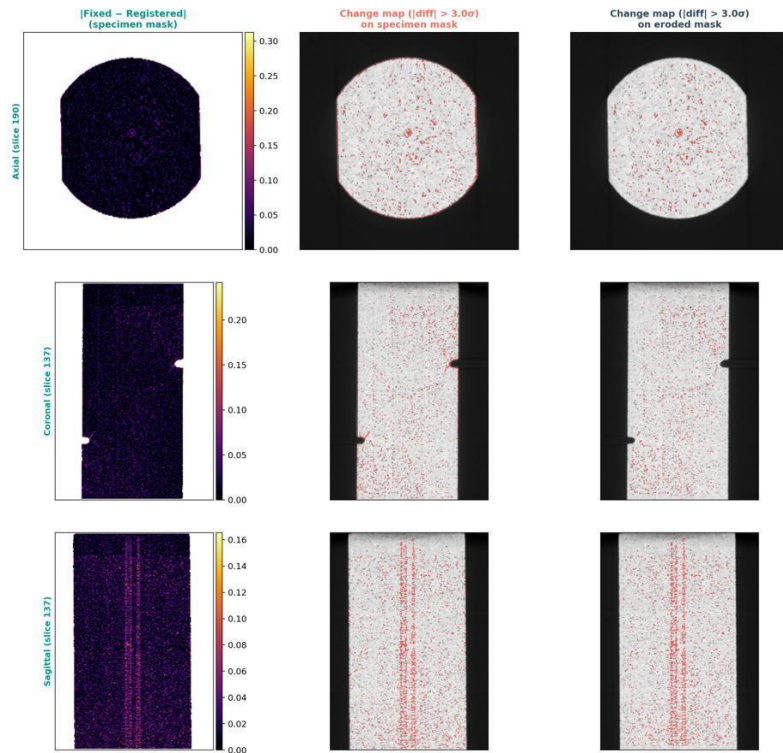


Distribution of Voxel Intensity Differences

- Narrower distribution after registration = most voxels now match well



Deformation Analysis Inside Specimen



Key Findings

Threshold

$3\sigma = 0.042$

Interior changed

5.78%

Full mask changed

6.65%

Components

186,200

Largest cluster

20,626 voxels

5.78% is an upper bound - includes imaging artefacts (ring/stripe effects), not only true deformation.



Conclusions

Rigid registration (Euler3D + MI + multi-resolution) successfully aligned both CT volumes

Corrected $\sim 3^\circ$ Z-rotation and ~ 2 -voxel translation - MSE reduced by 87%, NCC = 0.997

Remaining interior differences ($\leq 5.78\%$) represent an upper bound on deformation + artefacts

Masked & eroded evaluation ensures boundary-effect-free, defensible metric assessment